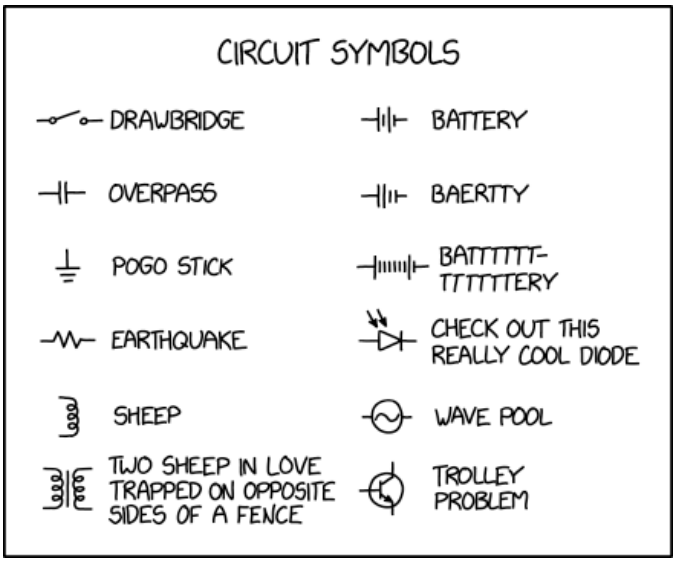
Circuit Lab Test

Science Olympiad Division C
Captain's Exchange Test 2025-26
Written By: Castro Valley High School



Names: _____

Team name and school: _____

Team Number: _____

MCQ: _____ / 51
FRQ: _____ / 45
Score: _____ / 96

Rank: _____

1. What experiment led Michael Faraday to discover electromagnetic induction? [1]
    a. **Faraday coiled 2 pieces of wire around an iron ring. When a current was passed through 1 coil, the other experienced current momentarily.**
    b. Faraday fired photons on a metal plate, and measured the ejected electrons from the plate.
    c. Faraday flew a kite in Philadelphia, which struck lightning and caused a spark with the key attached to it.
    d. Faraday built a device of alternating discs of zinc and copper separated by wet paper soaked in saltwater, which provided a stable source of electrical current.

2. What year did Nikola Tesla invent the Tesla Coil? [1]
    a. 1887
    b. 1891
    c. 1906
    d. 1913

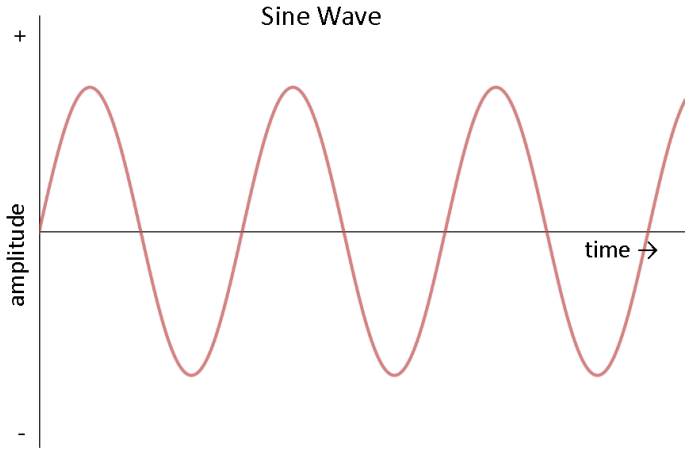
3. Which scientist is most well known for describing the relationship between current, resistance, and voltage? [1]
    a. Gustav Kirchhoff
    b. Werner von Siemens
    c. Georg Ohm
    d. Alessandro Volta

4. Which scientist invented the solenoid? [1]
    a. Joseph Henry
    b. James Clerk Maxwell
    c. Michael Faraday
    d. André-Marie Ampère

5. If a proton was placed in an empty electric field, where would the electric field lines point? [1]
    a. Towards the proton.
    b. Away from the proton.
    c. The field lines don't change.
    d. Field lines only change if the proton is moving.

6. If the distance between 2 charges doubles, what happens to the electric force? [1]
    a. It becomes ¼ as strong.
    b. It becomes ½ as strong.
    c. It becomes 2x as strong.
    d. It becomes 4x as strong.

7. What are applications that use Direct Current (DC)? [1]
    a. Refrigerators, Washing Machines, HVAC
    b. Electric Trains, Power Lines, Transformers
    c. Cable Television, Induction Motors, Power Outlets
    d. Charging Phones, Car Batteries, Solar Panels



8. Does the figure above represent an AC graph or a DC graph? [1]
    a. DC
    b. AC

9. What is a potential hazard that Alternating Current (AC) has over Direct Current (DC)? [2]
    a. When shocked with AC, a person will undergo a series of muscle contractions compared to a continuous muscle contraction with DC.
    b. AC will send constant shocks to the body, while DC will send pulses of electricity.
    c. AC is always at a higher voltage than DC, leading to a higher current passing through the human body.
    d. AC only passes through the body if the person is wearing gloves made of conductive materials, where DC will pass if the gloves are made of insulating material.

10. What direction is the flow in conventional current? [1]
    a. Negative to Positive.
    b. The direction cannot be determined.
    c. The same as electron flow.
    d. Positive to negative.

11. In the modern day, what is the typical AC voltage of a house in the US? [1]
    a. 90V
    b. 120V
    c. 175V
    d. 240V

12. What are the 3 main causes of static electricity? [1]
    a. Friction, Magnetism, Climate
    b. Resistance, Aperture, Exposure
    c. Friction, Separation, Induction
    d. Current, Charge, Elasticity

13. Which of the following DECREASES the electrical resistance of the human body? [1]
    a. Wearing insulated gloves
    b. Wetting your hands
    c. Providing a smaller surface area for contact
    d. Standing in a dry environment

14. Two electrons are a distance 0.5 cm apart. What is the electromagnetic force exerted by one electron on the other? [2]
    a. 9.2×10^−24 N
    b. 3.8x10^-15 N
    c. 1.4x10^-7 N
    d. 2.1x10^-5 N

15. 4 capacitors are in series, with their respective capacitances being 1μF, 3μF, 2μF, and 3μF. What is the total capacitance of the system? [3]
    a. ½ μF
    b. 6/13 μF
    c. 6/11 μF
    d. 9 μF

16. What is generally considered a lethal current of the human body? [1]
    a. 0.2A
    b. 0.5A
    c. 2A
    d. 4A

17. If a parallel-plate capacitor had a capacitance of 10^-8 Farads and the area of each plate is 30cm^2, what distance must you separate the plates? [2]
    a. 1.8x10^-7m
    b. 2.6×10^−6m
    c. 1.1x10^-5m
    d. 2.3x10^-4m

18. An MP3 player is typically powered by four 1.5V batteries, wired in Series. How much resistance is in the MP3 player if it pulls a current of 0.02A? [1]
    a. 100 Ohms.
    b. 150 Ohms.
    c. 300 Ohms.
    d. 850 Ohms.

19. What are the 3 terminals in a 3 way light switch circuit? [1]
    a. 1 traveler terminal, 2 common terminals
    b. 1 common terminal, 2 traveler terminals
    c. 1 common terminal, 1 traveler terminal, 1 central terminal
    d. 3 ground terminals

20. What does the last color on a 6 band resistor indicate? [1]
    a. Multiplier
    b. 3rd Digit
    c. Temperature Coefficient
    d. Tolerance

21. What is the region known as where electrons and holes are absent in a PN junction? [2]
    a. The Diffusion Border.
    b. NP Equilibrium.
    c. Free Space Junction.
    d. **The Depletion Layer.**

22. Which of the following is NOT an application associated with Operational Amplifiers (Op-Amps)? [2]
    a. As a Transformer
    b. As a Differentiator
    c. As a Filter
    d. As a Phase Shifter

23. A current carrying wire has a current of 6A running through it. What is the magnitude of the magnetic field from 2 cm? [3]
    a. 3.0x10^-5 T
    b. 6.0×10^−5 T
    c. 1.2x10^-4 T
    d. 6.7x10^3 T

24. If a transformer's primary coil has 20 turns and the secondary coil has 30 turns, what is the output voltage when 15V is sent through? [1]
    a. 17.5V
    b. 22.5V
    c. 25V
    d. 31.5V

25. Which of the following is commonly considered a semiconductor? [2]
    a. Lead
    b. Germanium
    c. Gold
    d. Hassium

26. What process allows Light Emitting Diodes (LEDs) to produce light? [2]
    a. Piezoelectricity
    b. Electroluminescence
    c. Corona Effect
    d. Photoconductivity

27. What are the characteristics of components in a series circuit? [1]
    a. Voltage is divided, current is constant between components.
    b. Voltage is constant, current is divided between components.
    c. Both Voltage and Current are constant between components.
    d. Components act the same, regardless of wired in Series or Parallel

28. If you spread out the plates in a parallel-plate capacitor farther away from each other, what happens to the capacitance? [2]
    a. The total capacitance increases linearly.
    b. The total capacitance stays the same.
    c. The total capacitance decreases.
    d. The total capacitance increases exponentially.

29. Which of the following are considered “Universal Logic Gates”? [1]
    a. AND & OR
    b. NOR & NOT
    c. XNOR & OR
    d. NAND & NOR

30. What is the transfer of electrons and holes across a PN junction known as? [2]
    a. Diffusion
    b. Stimulated Amplification
    c. Depletion
    d. Rectification

31. What is TRUE about Non-Inverting Operational Amplifiers? (Op-Amps) [3]
    a. Their gain is always greater than 1
    b. Output is 180 deg out of phase with input signal
    c. Does not have the capability to act as a voltage follower
    d. Their positive terminal is connected to the ground

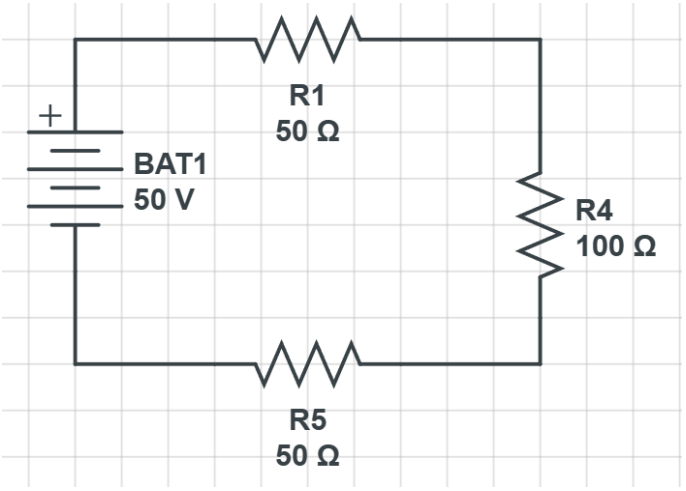
32. Which of the following are components of a DC motor?[1]
    a. Rectifier, Amplifier, Transformer
    b. Commutator, Shaft, Armature
    c. Inductor, Resistor, Capacitor
    d. Controller, Nacelle, Pitch

33. What is the difference between a high-pass filter and a low-pass filter? [2]
    a. Low-pass allows high-frequency signals to pass while reducing low-frequency signals, and High-pass does the opposite.
    b. High-pass attenuates all signals, while low-pass amplifies all signals.
    c. High-pass allows high-frequency signals to pass while reducing low-frequency signals, and low-pass does the opposite.
    d. Both filters operate exactly the same.

34. What electrical component would most likely be found in an Electrocardiogram (ECG), an analog device measuring electrical signals of the heart? [2]
    a. Operational Amplifiers (Op-Amp)
    b. MOSFETs
    c. Inductors
    d. Oscillators

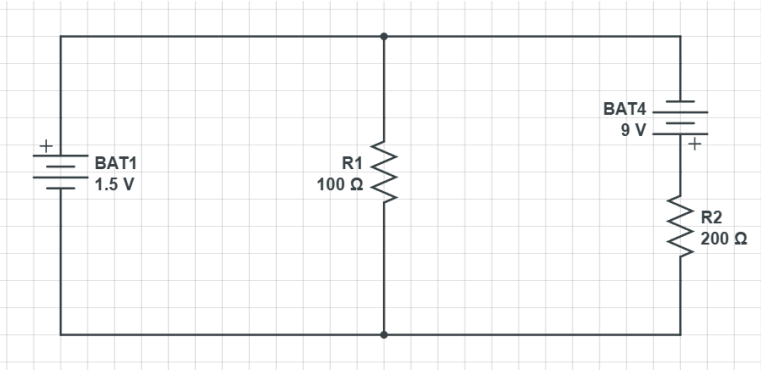
FRQ Section:

35. What is the current across this circuit? [1]



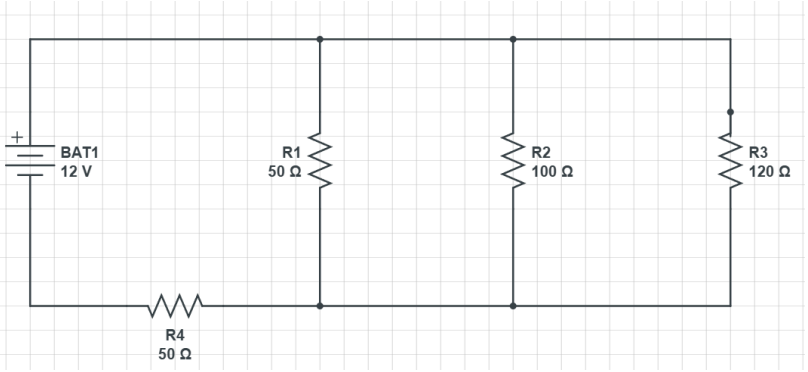
+1 for Answer: 0.25A

36. What is the current passing through R2 in this circuit, and in what direction? [3]



+1 for setting up proper junction equations ($9-200I_{R2} + 100I_{R1}$)
+2 for answer: 0.0525A going down

37. How much power is being dissipated in this circuit? [3]



+1 for parallel resistance
+1 for finding total resistance
+1 for using P = VI to find answer: 1.89 Watts

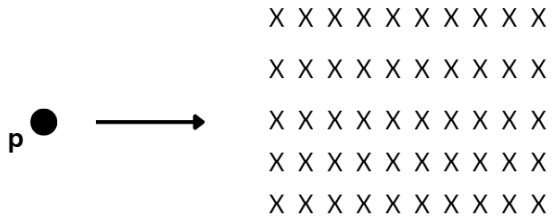
38. A hydrogen atom has a mass of 3.2x10^-27kg. If the hydrogen atom is sent into a chamber at 10000m/s where the magnetic field points at a strength of 0.4T perpendicular to the atom, calculate the diameter of the circular path that the hydrogen atom makes. [4]

+2 for using equations F = qvb and F_c = mv^2/r
+1 Plugging in info to find radius
+1 for answer: 0.001m (in scientific notation is fine)

39. Which force is stronger? The electric force or the magnetic force between you or the chair you are (presumably) sitting on? Explain. [3]

+1 for mentioning the electric force is bigger than the magnetic force
+1 for magnetic force only acting on moving charges, while electric acts on all charges, like the charges from body to chair
+1 for the $F_{electric}$ constant being significantly bigger than $F_{magnetic}$

40. A proton is being fired at 140 m/s into a uniform magnetic field with the strength of 0.3T. What is the magnitude of the magnetic force, and in what direction? [4]



+2 for finding magnitude of force using F = qvb. Answer: 6.72x10^-18 N
+2 for finding direction of the force: Upwards

41. What is the property that makes certain materials far better insulators than others? Briefly Explain. [2]

+1 for electrical resistivity
+1 for insulators' electrons are tightly bound to their atoms, preventing energy transfers

42. Explain how a transformer changes the voltage of electricity, and why does it only work with Alternating Current (AC)? [3]

+2 for electromagnetic induction, with a changing magnetic field inducing a voltage from 1 coil to another
+1 for AC producing magnetic flux, while DC doesn't, allowing induction to occur.

43. A single lemon battery produces about 0.7V of electricity. If a washing machine requires 15A to operate and the circuit has a 10K Ohm resistor connected, how many lemon batteries are required to be wired in series to operate the washing machine? [2]

+1 for using Ohms law for 1 lemon: 7.0 x 10^-5 A
+1 for determining how many lemons are required: 214,286 lemons

44. How does electromagnetic induction generate enough heat for induction cooktops to function? [3]

+1 for AC generating magnetic flux, allowing induction to occur
+2 for eddy currents being present in the metal pan, which heats up the metal

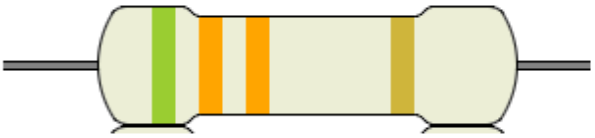
45. 1200 loops of a wire creates a solenoid with a length of .6m, carrying 5A and with a radius of 60cm. What is the magnetic field at the center of the solenoid? [3]

+2 for using Solenoid formula and plugging values in: B = μ0 * n * I
+1 for finding the answer: 0.0126T

46. The power cable to your Samsung Smart-Fridge has a current of 90 μA flowing through. How long does it take for a charge of 25 Coulomb to pass through the wire? Answer to the nearest hour. [2]

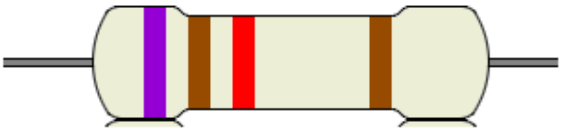
+1 for using I = Q/t
+1 for finding the answer: 77 hours

47. What is the resistance of this resistor?[2]



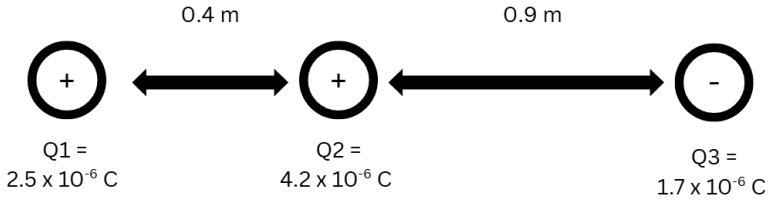
+1 for Answer: 53 k ohms, 5% tolerance

48. If 2 of these resistors were wired in series, what would be the total resistance of the circuit? [3]



+2 for finding the resistance of 1: 7.1K ohms
+1 for 2 resistors in series: 14.2K ohms

49. What is the magnitude & direction of the net force of the Q2 particle? [3]



+1 for using Coulombs Law ($F = k * (q_1 * q_2) / r^2$)
+1 for finding the net force: 0.6692N
+1 for finding the direction of the force: towards the right

50. Some tasers online have been advertised as delivering 7 million volts across two prongs that are only 1 cm apart. Why is this design not feasible? (In other words, why would the taser not be able to deliver the advertised voltage?) [4]

+4 for the voltage exceeding the dielectric strength of air, which breaks down at 30kV and would short circuit the taser.